

AWG / DDS Function Generator

Engineering Process Report

How we got here

Current build: **v1.2** — working, characterised

Three PCB revisions; two arrived dead

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Compiled from the timestamped engineering logs, the documentation set, and the current state of the repository. Figures and dates are cited to their source throughout; see [About this report](#) for what is and isn't drawn from the logs.

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Summary

0.1 Why?

This project was aimed to be above all else, a learning experience with a byproduct of a working function generator. It was an excuse work with digital analog synthesis, signals, filters, opamps and even switches. A gateway to mixing analog and digital, build intuition and experience. Along with that, having a set goal, a tool for proof helped with rough prototyping days, as well as motivation.

0.2 How?

The idea begun with an arbitrary wave generator. It's in theory, a generator that can output any wave, ideally at any frequency with unlimited power. With as a template, after online research the goal of 1MHz was established, using a raspberry pi pico W. The pi pico outputs using an R-2R ladder "steps", that is samples of the requested signal, and using the proper filters and reconstruction methods, the full signal is recovered. Ideally the project would follow a R-2R -> Reconstruction -> Buffer -> output diagram, which it mostly did with the exception of a variable gain stage, as the pico is limited to +3.3Vp-p. Moreover, due to the lack of a proper power supply, it was designed around 2x9V rechargeable batteries.

0.3 The Arbitrary Wave Generator (AWG)

The instrument itself is a laboratory-oriented DDS/AWG signal generator built around an R-2R ladder, a 5th-order doubly-terminated LC Butterworth reconstruction filter, and a Raspberry Pi Pico W. It uses lm318 at 3 stages, during reconstruction, amplification and buffering, as well as an TC4427 for squarewave buffering and an ADG1419 for output selection. It produces sine, square, noise, AM and FM under USB control, it is characterised from DC to 1MHz on the sine path and to at least 2MHz on the square path, and has a known, documented, firmware-mitigated fault in its filter that is described here rather than smoothed over.

1 About this report

This is a process report, not a design document. It exists to answer *how did we get here*, not to re-state what the instrument does — the documentation set already covers that, in `design-philosophy.md` for intent, `v1_2_architecture.md` for the current architecture, and `measurements.md` for numbers. This report's job is the timeline and the honesty around it: what was tried, what worked, what didn't, and what it cost.

Sourcing. Every dated event below is read from the timestamped engineering logs (18 files, 27 January – 1 September 2026) and cross-checked against `lessons.md`, `version_requirements.md` and `measurements.md` for figures the logs state only loosely. Where a claim comes from a log entry it is cited by filename, so it can be checked against the original. A complete index of those files is in the source-material appendix.

The logs contain no entry at all for v1.1's failure. That account (Section 3.5) was supplied directly by the author while reviewing this draft rather than found in the source material. It is flagged as such where it appears, and in the appendix, rather than presented as if it were logged like everything else here.

What this is not. It is not a substitute for `lessons.md` — the rules derived from failure live there, and this report narrates the failures that produced them. It is not a spec sheet either; that

is the top of README.md.

2 Project at a glance

Instrument summary — v1.2	
Waveforms	Sine, square, sine+noise, AM, FM
Sine range	0 Hz–1 MHz, characterised; DDS with phase accumulator
Square range	To at least 2 MHz
Sine amplitude	~16 V _{pp} max at 1 kHz, analog pot
Square edges	~20 ns rise and fall, 10–90%
Output impedance	53 Ω sine / 58 Ω square, measured at 1 kHz
Resolution	8-bit R-2R ladder, ~7 effective bits
Supply	USB and 2×9 V batteries
Control	USB CDC serial and desktop UI (scripts/awg_ui.py)
MCU	Raspberry Pi Pico W (RP2040), overclocked to 225 MHz

Full current numbers are in measurements.md; the full specification table is at the top of README.md.

Version	What changed	Outcome
v0.x	Breadboard. R-2R DAC, discrete and op-amp filtering, discrete square path.	Architecture validated
v1.0	First PCB. Mixed THT/SMD, 4× cascaded RC filter.	Dead on arrival — BOM
v1.1	All-SMD, LC Butterworth reconstruction filter.	Dead on arrival — schematic
v1.2	As v1.1, with the LM318 short fixed and the Pico soldered direct.	Working, characterised

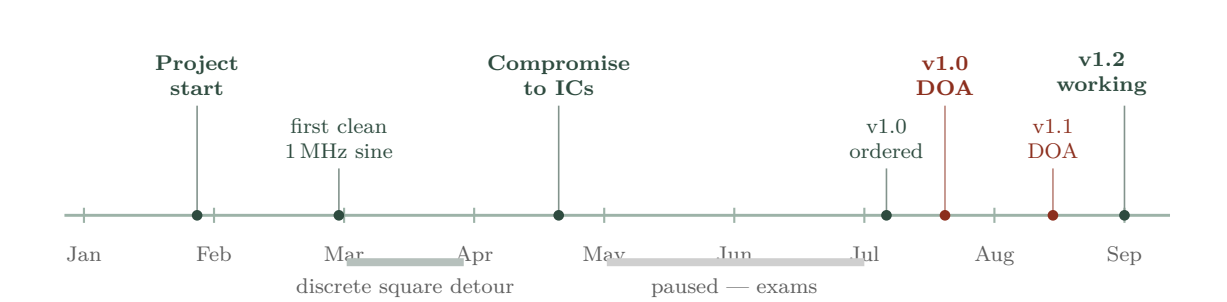


Figure 1: Eight months, January to September 2026. Two of the three PCB revisions arrived dead; the gap between them is exam season, not debugging.

3 Timeline

3.1 Breadboard era: R-2R, filtering, and a lot of noise

The project started from nothing on 27 January 2026: Pico toolchain installed, Zadig drivers for USB, and a first, crude R-2R ladder built from 10 kΩ/20 kΩ resistors, driven by AI-generated test firmware for speed. Quality visibly degraded past 60–70 kHz and the ladder was usable to about 200 kHz; pushed hard, the bare Pico GPIO could toggle at up to ~10 MHz, which set an early — and, in hindsight, over-optimistic — sense of headroom.

Source: EngLog 27-01-26 - 0324.txt

By 18 February the ladder had been changed to $2.2\text{ k}\Omega/1.1\text{ k}\Omega$ (two $2.2\text{ k}\Omega$ in parallel) for better linearity, and serial frequency control was working end to end after a rewrite of the host I/O code. 1 MHz looked reachable, by the log's own hedged wording ("seems that I can manage"), but nothing had been done to condition or verify that signal yet: the buffer and filter parts it would take to find out were still on the order list.

Source: EngLog 18-02-26 - 1750.txt

The next entry is the first real filtering attempt: a TL082 unity buffer straight off the ladder, followed by two RC low-pass stages ($470\ \Omega/220\text{ pF}$) and a second buffer. Usable to 100–200 kHz; at 1 MHz the signal showed $\sim 50\%$ amplitude loss, roughly 1 V of DC offset, and visible jitter and ghosting. Square-wave firmware was added in parallel but not yet seriously characterised, pending a better op-amp.

Source: EngLog 19-02-26 2036.txt

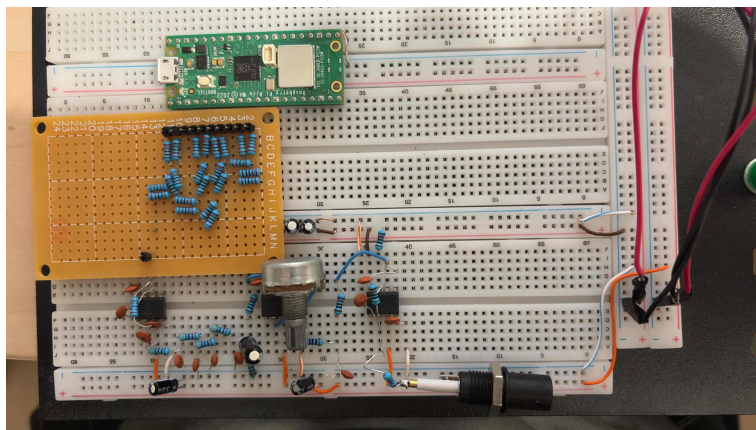


Figure 2: Breadboard sine chain during the RC-filter era: R-2R ladder, buffering and cascaded filter stages, before the move to LM318 and, later, to a PCB.

24 February added a high-pass stage immediately after the ladder to strip the DAC's DC bias, a $10\text{ k}\Omega$ potentiometer as a first amplitude control, decoupling ($100\text{ nF} + 10\ \mu\text{F}$) at the op-amp rails, and the first schematics committed to the repository. LM318 buffers, solid-core wire and a perfboard were ordered.

Source: EngLog 24-02-26 2100.txt

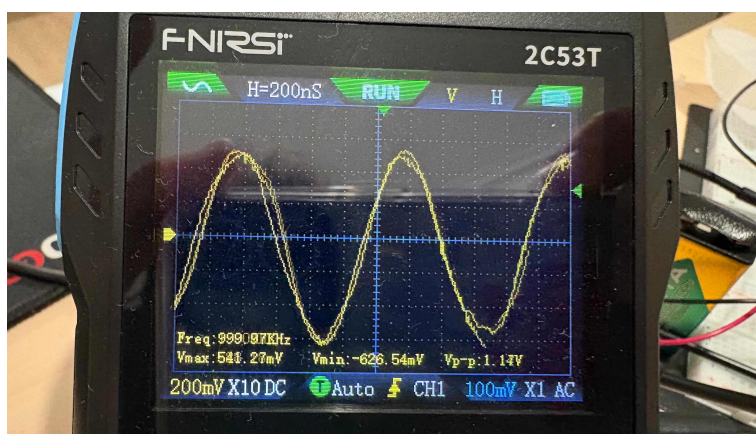


Figure 3: First quality check against a bench generator at 1 MHz, breadboard build.

The Sallen-Key detour. With LM318s in hand on 27 February, a full day went into active Sallen-Key filtering, using parallel resistor and capacitor networks to hit non-standard ratios. Results were persistently noisy, worse under load, and worse still through the gain stage. Four

candidate causes were ranked in the log itself — parasitic capacitance and poor breadboard contact as the top suspect, the unorthodox filter/buffer layout, bad R-2R contacts newly exposed by the better op-amp, and general wiring and grounding — with a closing line that reads, in hindsight, like the project correctly diagnosing its own next two failure modes:

“the brief moments of clean signal with each stage tells me its something less... systemic.”

Source: EngLog 27-02-26 0000.txt

First major success. The very next day, active filtering was abandoned entirely in favour of five cascaded passive RC stages (four $470\ \Omega/221\ \text{pF}$ low-pass, one $100\ \mu\text{F}/1\ \text{M}\Omega$ high-pass), a dedicated inverting gain stage ($1\ \text{k}\Omega$ input, $0\text{--}10\ \text{k}\Omega$ feedback pot, $5\ \text{pF}$ compensation) for rail-to-rail swing, a further low-pass stage to clean up noise the gain stage had uncovered, a voltage-follower output buffer, and a $47\ \Omega$ series resistor into a BNC connector — effectively the topology that would carry through to the PCB. The result was clean, with no ghosting and only a slight “vibration” at $1\ \text{MHz}$ that the log is honest about not being able to attribute to the circuit rather than the oscilloscope.

Source: EngLog 28-02-26 0003.txt

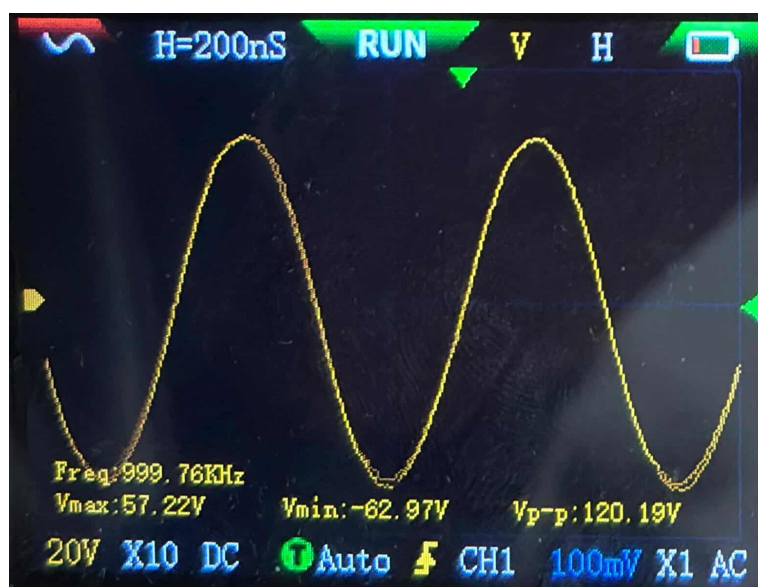


Figure 4: 1 MHz sine capture from the breadboard build, taken around the first major success (28 February 2026).

3.2 The discrete square-wave detour

Later the same day, breadboard contact problems fried an LM318 — a feedback loop opened under a loose contact. The decision was made to accept the sine chain’s Hi-Z-only, no-current-drive limitations as a deliberate trade for lower parasitics, add square-wave and arbitrary-waveform output, and move toward a first PCB.

Source: EngLog 28-02-26 1709.txt

What followed was roughly two weeks of discrete-transistor square-wave buffering, pursued specifically to build real experience with current amplification rather than reach for an IC immediately.

- **1 March.** Decision to go discrete; parts ordered. Source: EngLog 1-03-26 1637.txt
- **3 March.** Parts arrive. A hand-derived current amplifier, biased to remove the $\sim 0.7\ \text{V}$ junction drop, worked in SPICE but not on the bench. GPIO8 was burned by inserting a capacitor believed, incorrectly, to be discharged — a direct hardware casualty rather than a design flaw. Source: EngLog 3-3-26 2336.txt

- **13 March.** First working square-wave buffer prototype, limited to below ~ 50 kHz with rounded edges. Source: EngLog 13-03-26 2007.txt
- **25 March.** An unconventional totem-pole push-pull stage reaches a 1 MHz square wave — noisy, with a duty-cycle asymmetry the log calls “duty cycle variety”, and limited by a ~ 1 k Ω input impedance, visible heating in the transistors and resistors, and the battery’s own current capacity. Source: EngLog 25-03-2026 1739.txt
- **27 March.** The duty-cycle and frequency-mismatch bugs are traced to the first PNP pair’s 9 V rail, and removing it fixes the 1 MHz case — but only at high frequency; the fix did not carry down to the low-frequency case. Source: EngLog 27-03-26 0031.txt

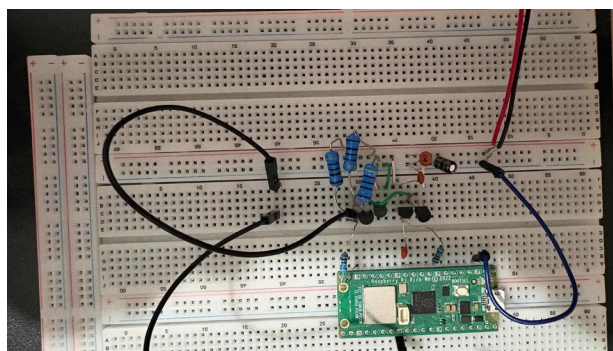


Figure 5: The discrete-transistor square-wave buffer on breadboard, from this detour — the circuit itself, not a captured output waveform.

3.3 Compromise: ICs, requirements, and a PCB

After the holidays, and after posting for help online without result, the log records a deliberate compromise: continuing the discrete-transistor path risked never shipping a working instrument. V1 requirements were fixed — sine and square from 0 to 1 MHz at 9 V peak, digitally adjustable amplitude, PC control via the Pico W, a durable PCB, strong documentation, BNC output, USB and battery power, and Hi-Z loads only — and the square path moved to a TC4427 gate driver, with amplitude control via adjustable LDO rail limiting (LT3080/LT3090) and a digital pot.

Parts for that scheme were ordered, but neither made it onto the PCB. Simulation showed the LDO rails degrading the square wave itself, and the digital pot turned out to be unavailable in the resistance range needed or in a DIP package — and would have added noise of its own even if it had been.

Source: EngLog 20-04-26 0103.txt

Project work was then paused through May and into June for exams. Source: EngLog 01-05-26 1714.txt

3.4 v1.0: first PCB, dead on arrival

The first PCB was designed and ordered as a mixed THT/SMD board — filtering kept through-hole for rework access, given how differently breadboard and PCB parasitics were expected to behave — using the four-stage cascaded RC filter from the breadboard build. The rail-limiting amplitude-control scheme from April was tried and dropped: it caused visible waveform rounding, so amplitude reverted to a simple variable resistor for this revision. Source: EngLog 6-7-26 1439.txt

The board arrived, and the failure was immediate and total. The BOM matcher had substituted 2.21 Ω resistors for 2.21 k Ω across the entire R-2R ladder: three orders of magnitude off, on every rung. This is Lesson 1 in Section 6.

Source: EngLog 19-07-26 1656.txt

The same log entry records the immediate pivot: send the board for a respin, replace the four cascaded RC stages with a 5th-order doubly-terminated LC Butterworth filter (~ 1.5 MHz cutoff, C0G capacitors, 0603 inductors), and move to a fully SMD, JLCPCB-assembled board. The RC cascade was already a known amplitude bottleneck by this point — each stage ate into the signal well before its own corner — and researching filter topologies at the time made the Butterworth's flatter passband and steeper rolloff the obvious replacement, independent of the BOM failure that forced the respin anyway.



Figure 6: v1.0 (left) and v1.1 (right) side by side — the two boards that arrived dead. v1.0's $2.21\ \Omega$ ladder and cascaded-RC filter section are visible at top left; v1.1 already carries the LC Butterworth (L1/L2), the soldered-down Pico and the panel-mount BNC that would carry through to v1.2.

3.5 v1.1: Butterworth redesign, dead on arrival

This version arrived, was soldered up, and went straight to testing — to no output at all. The ladder's test point showed nothing and the board was completely silent, with no immediate way to tell whether the cause was power, firmware or a bad contact.

Pivoting to the square path as a diagnostic step paid off: it worked, and worked well, with only some edge ringing that turned out to be a probing artefact — it disappeared once the scope ground lead was shortened — rather than a property of the signal. With sine dead and square clean, the fault was narrowed to the sine chain specifically.

Going over the board repeatedly turned up the cause: a stray trace, left behind during the KiCad-to-Altium migration, shorting the two inputs of the first-stage LM318. The board's layout, combined with not having a hot-air rework station on hand, made it impractical to confirm that this was the *only* fault by reworking and re-testing in place. After signal insertion at the DAC test point, the rest of the board proved functional at the lower frequencies the available handheld generator could produce. With the schematic fix in place, v1.2 was ordered directly.

No log entry documents this failure or its diagnosis. The account above was supplied directly by the author, and `lessons.md #3` is the only other place it is summarised.

3.6 v1.2: working build

With the LM318 short fixed, power LEDs added and costs trimmed in several places, the board came up clean: tested before assembly, then the Pico soldered directly on. A desktop host

UI was added, along with noise, AM and FM modes on the firmware side. The log calls the project “officially completed” at this point, with bugfixes, additional firmware features and full characterisation flagged as the next phase rather than as open requirements. Source: EngLog

01-09026 0306.txt

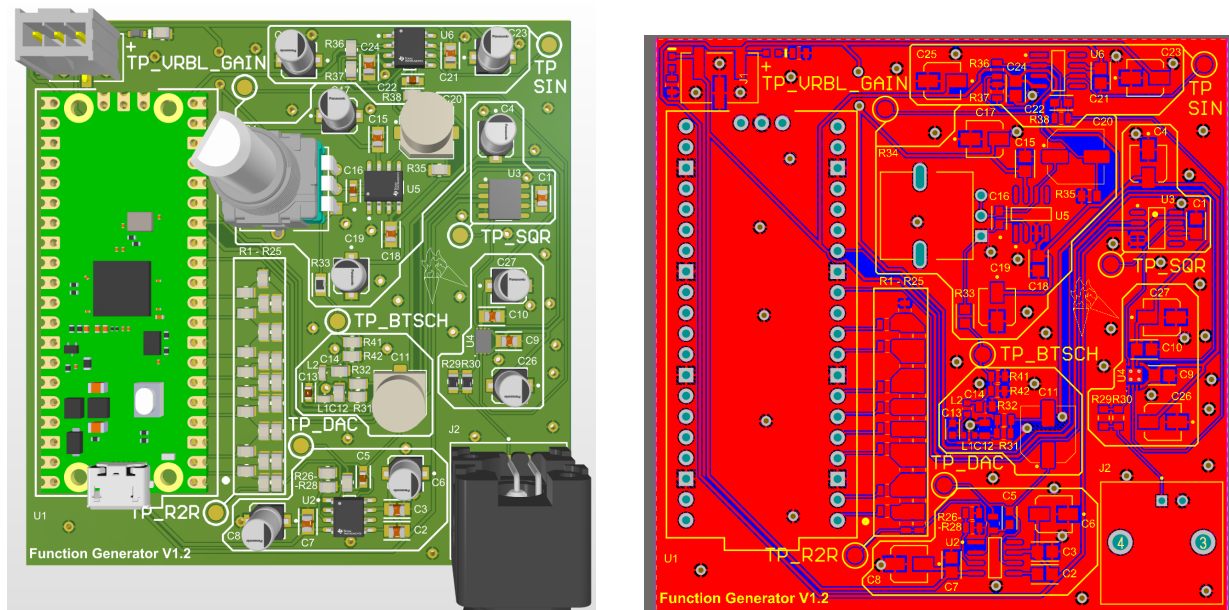


Figure 7: v1.2 as designed: Altium 3D render and copper layout, August 2026 — the board that was ordered.

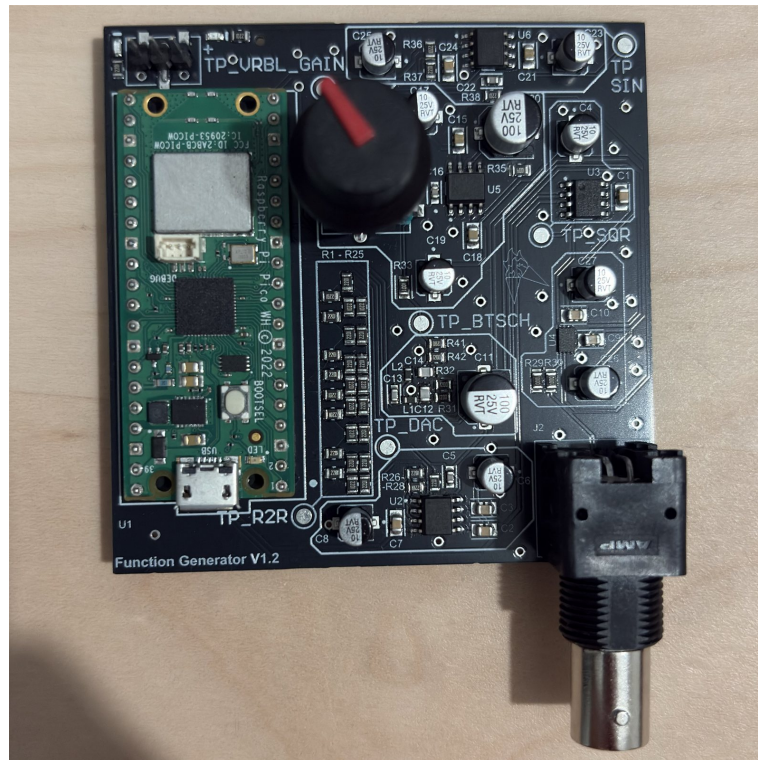


Figure 8: v1.2 as built: the board that arrived, assembled, with the Pico soldered directly on and the gain pot and BNC hand-fitted.

The very next log entry is where characterisation starts paying off. Full-drive sine flatness mea-

measurements surface a -3.1 dB droop at 1 MHz , and scaling the DAC lookup table by four in firmware eliminates it entirely with no hardware change — which is what pins the cause on inductor saturation rather than on the filter design or op-amp bandwidth (Lesson 3, Section 6). It is diagnosed cleanly and, so far, patched only in firmware; the real fix is expected to be small, swapping L1/L2 for parts with adequate saturation current, likely reworkable on the existing board rather than the kind of fault that forces another respin. A firmware fix for the square wave's duty-cycle asymmetry is logged in the same entry. Source: EngLog 01-09-2026 1436.txt

4 What went wrong

- **Every failure that cost a board was invisible to every check that was not a physical measurement.** A BOM matcher dropping a unit prefix (Lesson 1), a copy-paste merging two nets across an EDA migration (Lesson 2), and a datasheet column never read (Lesson 3). None was caught by ERC, DRC, simulation or visual schematic review. A fourth, closely related near miss — name and part-number drift in the component library, folded into Lesson 1 alongside the BOM matcher — was caught early enough to cost nothing, which is the only reason it reads as a near miss rather than a fourth incident.
- **The Butterworth filter's inductors were selected on inductance, package, tolerance and self-resonance — and not on current rating** (Lesson 3), in a $60\ \Omega$ system where full-scale drive pushes real current through them. No simulation would have surfaced it: LTspice's ideal inductor has no saturation current to violate.
- **Real hardware casualties from real mistakes.** An LM318 fried by a loose breadboard contact opening its feedback loop, and GPIO8 burned by inserting a capacitor assumed, wrongly, to be discharged (both Section 3.2). Both are logged plainly, in keeping with this project's own standard for honesty.
- **Two weeks on a discrete square-wave buffer that gave way to an IC anyway.** Not a wasted two weeks — Section 5 credits the experience — but the log's own tone by late March ("I have realised I grew tired and I'm in way too deep") is a real signal that the discrete-only constraint was fought past the point of being useful, and should have been reconsidered sooner.
- **A public request for feedback that produced nothing usable**, noted once and not revisited — worth knowing before trying that channel the same way again.
- **No hot-air rework station on hand when v1.1's LM318 short was found** (Section 3.5). With the fault physically identified, the board's layout and the lack of that one tool made it impractical to confirm it was the *only* fault by reworking in place, so a full respin was ordered on a diagnosis that could not be closed out on the bench first.

5 What went right

- **Killing active filtering when it wasn't working, rather than debugging it further.** The Sallen-Key detour (Section 3.1) cost real time, but the decision to drop it for passive RC, and later passive LC, rather than chase parasitics in an already-suspect breadboard layout, is what produced clean signal the very next day.
- **Diagnosing by changing one variable, twice, at the points that mattered most.** The 1 MHz sine droop was pinned on inductor saturation by scaling the DAC table alone and watching the droop disappear with no hardware change (Section 3.6); the discrete square buffer's duty and frequency bug was pinned on the first PNP pair's rail by removing exactly

that (Section 3.2). Both are textbook single-variable changes, and both are why the causes in this report are stated with confidence rather than hedged.

- **Choosing a passive LC Butterworth over stacking more active stages** once the RC cascade's amplitude loss became the limiting factor — no added op-amp bandwidth limit, no added noise, and a much steeper rolloff for the same board complexity.
- **The compromise to use ICs for square-wave buffering.** Two weeks of discrete-transistor work (Section 3.2) built real understanding of current amplification, but the explicit decision to stop and use a TC4427 once shipping was at risk is why the project has a working square path at all.
- **Pivoting to the square path as a diagnostic move on v1.1** rather than continuing to poke at a completely silent sine chain. A working square output with no sine output narrowed the fault to one half of the board immediately, and is what made the stray shorted trace findable by review at all (Section 3.5).
- **Soldering the Pico directly to v1.2**, trading reflash convenience — now USB-only — for the durability a socket does not give a board meant to survive being used rather than demonstrated once.
- **Treating “it compiles clean” as insufficient evidence.** Both PCB failures passed every non-physical check available at the time; the rules derived from them (Section 6) are specifically about closing that gap on the next board, not about avoiding the tools that let the errors through.

6 Lessons and rules derived

Reproduced from `lessons.md`, which remains the canonical source — update it there first if anything here goes stale.

1. **BOM and part-identity errors.** Two incidents, same failure class: a part gets ordered as something other than what the library says it is, and nothing downstream of the library catches it.

The 2.21 Ω incident. The BOM matcher substituted 2.21 Ω for 2.21 k Ω across the entire R-2R ladder. Cost: one full PCB respin (v1.0).

Name and number drift. Separately, a library entry named as a 0.1% thin-film part carried the LCSC number of a 1% thick-film part. Harmless in that position; the same drift in a precision position would have silently broken the design. Caught before it cost anything, which is the only reason it reads as a near miss rather than a second respin.

- Read the matched part *description* of every BOM line before paying — value and unit.
- Part numbers live at the library level, not per order. The library prevents the error class; the review-page read catches the rest.
- Library name, description and LCSC number must agree with each other.
- Audit periodically via a parameter-grid view — inconsistencies are visible in a column that are invisible one component at a time.
- Copy the final chosen number back into the library the same day as any order-time substitution.

2. Migration errors. A KiCad-to-Altium migration merged two LM318 inputs onto a single net via a copy-paste that left no visible trace. ERC passed. Cost: one full PCB respin (v1.1).

- After any cross-EDA paste or import, run ERC *and* a cold continuity check between the input pins of every op-amp instance — not just the one being worked on.
- A schematic that compiles clean is not a schematic that is correct.

3. Maximum current rating. Butterworth filter inductors were selected on inductance, package, tolerance and self-resonance — not on current rating. Rated 3mA; actual peak current at full scale is ~ 15 mA. The saturating ferrite costs -3.1 dB at 1 MHz where the design predicts a flat response.

- For series elements in a low-impedance signal path, compute the actual current and check it against the rated and saturation current before selecting — explicitly, because a $60\ \Omega$ system turns volts into real milliamps in a way that is easy not to think about when reasoning in volts and hertz.
- Simulation will not catch this: LTspice inductors are ideal and current-independent.
- Hotfix: the firmware low-current mode (lookup table $\div 4$) restores linearity at the cost of amplitude and two bits of resolution — diagnostic-grade, not a fix.
- Real fix: the same inductance in a larger package with adequate saturation current.

7 Current state — v1.2

Full numbers live in `measurements.md` and at the top of `README.md`; this section is the short version, current as of this report.

Working. Sine from 0 Hz to 1 MHz, DDS with a phase accumulator. Square to at least 2 MHz, PIO-generated, with user-settable duty and a TC4427-asymmetry correction. Sine+noise (SNR 0–60 dB, nominal), AM (0–100% depth) and FM (deviation in Hz). USB CDC serial control with a plain-text command protocol and a desktop UI. No DC offset observed. Output impedance measured rather than assumed: $53\ \Omega$ sine, $58\ \Omega$ square.

Known limitations. Full detail and causes are in `limitations.md`. In short: the filter inductors are under-rated for the current they actually carry (Lesson 3, with a firmware workaround rather than a fix); the filter corner sits at ~ 1.0 MHz against a 1.5 MHz simulation and this is still unexplained; the 9 V-peak output requirement is not met on ± 9.5 V rails; the instrument drives Hi-Z loads only, with no output buffer after the switch; amplitude control is an analog pot rather than digital; and THD and jitter are uncharacterised for lack of equipment.

8 Open items and future work

Drawn from `version_requirements.md` and `future-work.md`.

- **Digital amplitude control** — firmware lookup-table scaling, possibly with coarse analog gain switching. High priority in `future-work.md`.
- **Noise, AM and FM are implemented and running** but still listed as a future requirement, and their key parameters (noise-mode SNR, AM depth) are calculated rather than bench-calibrated. Both should be resolved together rather than documented as two separate gaps.

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- **The DDS paths accept commanded frequencies well past their ~ 2.017 MHz Nyquist limit with no check**, and alias silently rather than clamp or error. This is exactly the mechanism behind the earlier “2 MHz sine” reading that turned out to be an alias.
 - **Square duty and its correction toggle have no control in the host UI** — serial-only for now.
 - **Output impedance is measured on one unit only**; board-to-board variation is untested.
 - **Deliberately deferred:** higher-voltage output, selectable output impedance, output protection, an onboard display, standalone operation and self-calibration all remain in `future-work.md`.

Appendix A: source material

Log file	Covers
EngLog 27-01-26 - 0324.txt	Project start; first R-2R ladder
EngLog 18-02-26 - 1750.txt	Serial I/O working; ladder resistor change
EngLog 19-02-26 2036.txt	TL082 buffer and filter; square wave added
EngLog 24-02-26 2100.txt	HPF, pot, decoupling; schematics started
EngLog 27-02-26 0000.txt	Sallen-Key detour; noise-cause ranking
EngLog 28-02-26 0003.txt	Passive filter redesign; first major success
EngLog 28-02-26 1709.txt	LM318 fried; decision to move to PCB
EngLog 1-03-26 1637.txt	Decision to go discrete for square wave
EngLog 3-3-26 2336.txt	Parts arrive; GPIO8 burned
EngLog 13-03-26 2007.txt	First discrete square buffer (<50 kHz)
EngLog 25-03-2026 1739.txt	1 MHz totem-pole square prototype
EngLog 27-03-26 0031.txt	Duty-cycle and frequency bug fixed
EngLog 20-04-26 0103.txt	Compromise to ICs; V1 requirements set
EngLog 01-05-26 1714.txt	Project paused for exams
EngLog 6-7-26 1439.txt	v1.0 PCB ordered
EngLog 19-07-26 1656.txt	v1.0 DOA ($2.21\ \Omega$); v1.1 re-order decided
<i>(author account, no log file)</i>	v1.1 DOA; LM318 short found and diagnosed
EngLog 01-09-26 0306.txt	v1.2 complete; UI, noise, AM, FM added
EngLog 01-09-2026 1436.txt	Inductor saturation found; duty cycle fixed

No log entry documents v1.1's own failure or its diagnosis. That account (Section 3.5) was supplied directly by the author, and `lessons.md #3` is the only other place it is summarised.

Appendix B: current schematic (v1.2)

Dated 26 August 2026, reproduced at full page width overleaf. The embed is vector — scroll or pinch-zoom reads every net and value losslessly — and the schematic is also a link: click it to open [hardware/schematic_overview.pdf](#) on GitHub, a real URL that works from anywhere this PDF is opened, phone included, unlike a relative path into a folder that may not travel with the file.

